



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2024-25(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

End Semester

Semester: V

Programme: B. Tech Information Technology

Course: Theory of Computation

Time: 3 Hour

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33502	Course Outcome
CO1	To illustrate finite state machines to solve problems in computing.
CO2	To classify theoretical foundations of computer science from the perspective of formal languages.
CO3	To familiarize Regular grammars and Context Free Grammars.
CO4	To enhance students' ability to understand and conduct mathematical proofs for computation and algorithms.
CO5	To explain hierarchy of problems arising in computer science.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define DFA.	CO1	2	2
b.	What is closure property?	CO2	2	2
c.	Define CFL.	CO3	2	2
d.	What is TM?	CO4	3	2
e.	Define Halting problem.	CO5	2	2
Group B				

Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain DFA with example.	CO1	2	5
b.	Convert ϵ -NFA to DFA.	CO1	2	5
c.	Explain Moore and Mealy machines.	CO1	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain pumping lemma.	CO2	2	5
b.	Convert RE to FA with some examples.	CO2	3	5
c.	Explain regular grammar.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain CFG and ambiguity.	CO3	2	5
b.	Convert CFG to CNF with some examples.	CO3	3	5
c.	Explain PDA.	CO3	3	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain TM and its variants.	CO4	2	5
b.	Explain recursive languages.	CO4	3	5
c.	Explain LBA.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain PCP.	CO5	2	5
b.	Explain Halting problem.	CO5	3	5
c.	Explain recursive functions.	CO5	3	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create